

Moaz Ahmed Foad

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SUMMARY

Aspiring Mechatronics Engineer with foundational training in embedded systems and control engineering. Completed a technical track that enhanced my problem-solving skills as part of a business start-up project. Seeking to leverage my technical and problem-solving skills in a dynamic engineering environment.

EXPERIENCE

Control Sub-Team Member

FITENTH Racing Team

September 2025 - Present, Cairo

- Developed and tuned autonomous navigation and control algorithms for the race vehicle using LiDAR-based perception data.
- Implemented Follow-The-Gap (FTG), Pure Pursuit, and Model Predictive Control (MPC) algorithms for obstacle avoidance and trajectory tracking.
- Designed line-following and path-tracking controllers integrated with localization and mapping pipelines.
- Collaborated with localization and perception sub teams to integrate real-time autonomous driving architecture in ROS2.
- Tuned controller parameters and evaluated vehicle behavior under dynamic racing scenarios and constrained environments.

INTERN

EgSA-Egyptian Space Agency

January 2025 - February 2025, New Administrative Capital

- Designed and verified FPGA-based embedded aerospace systems using VHDL, contributing to FSM architecture, hierarchical modules, and arithmetic logic components; collaborated with engineering teams on lab projects and documented results through technical reports and presentations.

INTERN

GB Academy

August 2024 - July 2025, Giza, Egypt

- gained hands-on experience in automotive systems, working on major car mechanisms, diagnosing issues, performing on-site maintenance including tire services, and assisting with troubleshooting and repair of mechanical components to ensure optimal functionality.

EDUCATION

Mechatronics Engineering

Ain Shams University • Cairo, Egypt • 2027 • 3.29

CERTIFICATIONS

Python for Everybody Specialization

University of Michigan • 2026

Prompt Engineering for ChatGPT

Vanderbilt University • 2026

Introduction to Programming with MATLAB

Vanderbilt University • 2025

FPGA & VHDL Workshop

EgSA-Egyptian Space Agency • 2025

OSHA Construction & Industry Course

Orascom Construction PLC • 2025

SKILLS

Embedded Systems: FPGA Prototyping, Logic Design, Control Systems Design

Programming: Python, VHDL, C++

Electronics: PCB Design, Circuit Simulation, Microcontroller Integration

Mechanical Design: 3D Modelling, CAD, Engineering Drawings, Mechanism Design

Tools: SolidWorks, Eagle, MATLAB, Fusion 360